Year 12 Maths - Force and motion

Similar questions - Forces, resultants and equilibrium

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A particle is in equilibrium under the action of the three forces $\mathbf{F}_1 = \left(p\mathbf{i} + 2q\mathbf{j}\right)$ N, $\mathbf{F}_2 = \left(3q\mathbf{i} - 5\mathbf{j}\right)$ N and $\mathbf{F}_3 = \left(-7\mathbf{i} + 9\mathbf{j}\right)$ N, where p and q are constants.

(a) Find the values of p and q. **[4]**

(b) Find the magnitude of $\mathbf{F}_1$ in exact form. **[2]**

(c) Explain why the resultant of $\mathbf{F}_1$ and $\mathbf{F}_2$ has the same magnitude as $\mathbf{F}_3$, and find this magnitude. **[3]**

2 Two horizontal forces of magnitudes 20 N and 15 N act at a point O along bearings 030° and 120° respectively.

(a) Find the magnitude and bearing of the resultant of the two forces. **[4]**

(b) A third horizontal force $\mathbf{F}$ is added so that the three forces are in equilibrium. State the magnitude of $\mathbf{F}$ and the bearing along which it acts. **[2]**

(c) The third force is removed and the particle, of mass 5 kg, is free to move on a smooth horizontal surface under the first two forces only. Find the magnitude of its acceleration. **[2]**

Question 1

(a) Resultant zero. **i**: $p+3q-7=0$. **j**: $2q-5+9=0$, so $q=-2$ and $p=13$

(b) $\mathbf{F}_1 = 13\mathbf{i}-4\mathbf{j}$, so $|\mathbf{F}_1| = \sqrt{169+16} = \sqrt{185}$

(c) Since all three forces sum to zero, $\mathbf{F}_1+\mathbf{F}_2 = -\mathbf{F}_3$, which has the same magnitude as $\mathbf{F}_3$. $\mathbf{F}_1+\mathbf{F}_2 = 7\mathbf{i}-9\mathbf{j}$ and $|\mathbf{F}_3| = \sqrt{49+81} = \sqrt{130} = 11.4$ N

Question 2

(a) The bearings differ by 90°, so $R = \sqrt{20^2+15^2} = 25$ N. The angle between 20 N and R is $\tan^{-1}\frac{15}{20} = 36.9°$, so the bearing is $030° + 36.9° = 066.9°$

(b) 25 N on the opposite bearing, 246.9°

(c) $a = \dfrac{F}{m} = \dfrac{25}{5} = 5\ \mathrm{m\,s^{-2}}$